

AIPD Rule Book v1.0

Adaptive Interview Policy Dataset (AIPD)

Objective: Predict the Next Interview Policy using Semantic Features, Behavioral Features, and Interview Context Features.

Input Features

Category	Features
Semantic	Correctness, Concept Coverage, Reasoning Score, Missing Concepts
Behavioral	Engagement, Confidence, Hesitation, Eye Contact
Interview Context	Current Difficulty, Correct Streak, Wrong Streak
Output	Next Interview Policy (7 Classes)

Output Policies

- 1. Increase Difficulty
- 2. Maintain Difficulty
- 3. Reduce Difficulty
- 4. Probe Missing Concept
- 5. Ask Application Question
- 6. Ask Follow-up Question
- 7. Switch Topic

Rule Priority (First Matching Rule Wins)

Rule	Condition	Policy
Rule 1	Difficulty=Hard, Correctness $\geq$ 90, Coverage $\geq$ 88, Reasoning $\geq$ 88, Missing Concepts $\geq$ 4, Correct Streak $\geq$ 3	Switch Topic
Rule 2	Correctness<45 OR Wrong Streak $\geq$ 3	Reduce Difficulty
Rule 3	Missing Concepts $\geq$ 4 AND Correctness $\geq$ 50	Probe Missing Concept
Rule 4	Correctness $\geq$ 80, Coverage $\geq$ 80, Reasoning $\geq$ 80, Missing $\leq$ 2, Difficulty=Hard	Ask Application Question
Rule 5	Correctness 70-85, Coverage 65-80, Reasoning $\geq$ 80, Missing $\leq$ 3	Ask Follow-up Question
Rule 6	Difficulty=Easy, Correctness $\geq$ 90, Coverage $\geq$ 85, Correct Streak $\geq$ 2	Increase Difficulty
Rule 7	Otherwise	Maintain Difficulty

Behavioral Constraints

Excellent Candidate: Engagement 0.80-0.98, Confidence 0.80-0.98, Hesitation 0.05-0.20, Eye Contact 0.80-0.98

Good Candidate: Engagement 0.65-0.90, Confidence 0.60-0.85, Hesitation 0.15-0.40, Eye Contact 0.65-0.90

Average Candidate: Engagement 0.45–0.70, Confidence 0.45–0.70, Hesitation 0.35–0.60, Eye Contact 0.45–0.70

Poor Candidate: Engagement 0.20–0.55, Confidence 0.20–0.50, Hesitation 0.60–0.95, Eye Contact 0.20–0.55

Validation Rules

- Reject if Correctness>95 and Missing Concepts>3.
- Reject if Confidence>0.95 and Hesitation>0.80.
- Reject if Correctness<30 and Confidence>0.90.
- Reject if Correct Streak>0 and Wrong Streak>0.
- Reject if Difficulty=Easy and Policy=Reduce Difficulty.
- Reject if Difficulty=Hard and Policy=Increase Difficulty.

Suggested Dataset Distribution (100,000 Samples)

Policy	Samples
Increase Difficulty	15,000
Maintain Difficulty	15,000
Reduce Difficulty	15,000
Probe Missing Concept	15,000
Ask Application Question	15,000
Ask Follow-up Question	15,000
Switch Topic	10,000